HABITPULSEAI

AI-Integrated Habit Tracking and Personal Wellness System

PROJECT REPORT

A web application for managing daily routines, tracking habit completion, reviewing progress, and receiving AI-assisted habit guidance.

Report date: 26 October 2026

Prepared from the implemented application interface and project implementation notes

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About this report: This report documents the design, main features, data handling, visible application pages, and verification status of HabitPulseAI.

1. Abstract

HabitPulseAI is a web-based habit tracking and personal wellness application designed to help users build consistent routines through daily tracking, progress visualisation, and AI-assisted guidance. The application provides account registration and login, a dashboard, habit management, a calendar, analytics, an AI Coach interface, and profile settings. The interface presents a simple navigation structure and uses habit categories, emoji indicators, status labels, progress summaries, and motivational messages to make routine management easier to understand.

The implementation uses a React and Vite client with an Express and Mongoose server. Existing authentication and habit-management functionality are retained while enhancements improve responsive navigation, motivational quote presentation, recent activity tracking, optional suggested habit templates, habit-card details, and calendar presentation. Activity events are associated with the authenticated user and can be stored persistently when MongoDB is configured. Automated client and server test suites and a client production build were reported as passing during project verification.

The goal of HabitPulseAI is to support practical habit formation by making progress visible, reducing friction when adding or updating routines, and offering supportive guidance without requiring users to manage their habits through separate tools.

2. Introduction

Maintaining a routine requires regular action, a clear view of progress, and a manageable way to respond to missed days. A digital habit tracker can organise these activities in one place. HabitPulseAI addresses this need by combining habit creation and completion tracking with calendar-based review, analytics, and a conversational AI Coach interface.

The application is organised around a dashboard that summarises current habits and daily progress. Users can manage individual habits, inspect dates in a calendar, review completion patterns, ask habit-related questions, and update personal settings. The design uses a light interface with blue primary actions, lavender accents, rounded panels, and clearly labelled controls.

3. Project Overview

HabitPulseAI is a full-stack web application with a browser-based user interface and a server-side API. The client is implemented using React and Vite. The backend uses Node.js with Express, and Mongoose is used for MongoDB data modelling. The project includes authentication, user-specific habit records, analytics endpoints, and AI Coach interaction. The screenshots show the local application running at localhost:5174.

The main user journey is summarised below. The dashboard and My Habits page are designed to make routine actions accessible without navigating through multiple separate tools.

1. Create an account or log in
2. Review the dashboard
3. Add or select habits
4. Mark habits as completed
5. Inspect calendar and analytics views
6. Ask the AI Coach for guidance
7. Manage profile preferences

4. Objectives

- Provide account registration and login so users can access their own habit-tracking workspace.
- Allow users to create, edit, pause, resume, complete, undo, and delete habits.
- Summarise active habits, daily completions, completion percentage, streak information, and recent activity on the dashboard.
- Offer a calendar view for reviewing dates and completion records.
- Present analytics, including completion trends, habit mix, and habit performance.
- Provide an AI Coach interface with suggested questions about consistency, motivation, routines, and setbacks.
- Improve usability through optional habit templates, emoji and accent selection, responsive layouts, keyboard-accessible dialogs, and motivational messages.
- Keep user activity history associated with the authenticated account and support persistence through MongoDB.

5. Technology Stack and System Structure

Layer	Technology / Role
Frontend	React for component-based interface development; Vite for local development and production builds.
Backend	Node.js and Express for server-side routes and API handling.
Data layer	MongoDB with Mongoose models for application records, including habits and activity events.
AI Coach	AI Coach user interface for habit-related questions; the existing backend AI configuration is retained.
Testing	Vitest-based client and server tests, plus a Vite production build.

At a high level, the browser interface sends requests to the Express API. The server validates requests, applies authentication and user ownership rules, and reads or updates application records through Mongoose. The client then displays the resulting data in dashboard cards, habit cards, calendar cells, analytics charts, and activity lists.

6. Module Description

6.1 Landing page

Introduces HabitPulseAI with a short value statement, account creation and login actions, and a visual example of habit completion progress.

See Figure 1 in Section 8.

6.2 Registration and login

Provides forms for creating an account and signing in. The registration interface includes full name, email, password, and timezone fields. The login page requests email and password.

See Figures 2 and 3 in Section 8.

6.3 Dashboard

Displays a welcome panel, quick-add action, active-habit count, completed-today count, daily completion percentage, current streak, a completions-by-habit visualisation, today’s checklist, and recent activity.

See Figure 4 in Section 8.

6.4 My Habits

Shows optional suggested habit templates grouped by categories such as Health, Fitness, Learning, Productivity, and Mindfulness. Templates are only added after the user selects Add Habit. Habit cards show an emoji, category, description, frequency or target, completion status, streak, and actions for completion, editing, pausing, and deletion.

See Figures 5 and 6 in Section 8.

6.5 Calendar

Provides a month-based calendar with Previous Month, Today, and Next Month controls. Date cells can be used to review completion information; the implementation supports emoji indicators for completed habits.

See Figure 7 in Section 8.

6.6 Analytics

Presents a completion trend chart, a habit-mix chart, and habit-performance visualisation. The screenshots show the charts with zero completion values for the displayed sample period; this reflects the visible application state at capture time.

See Figure 8 in Section 8.

6.7 AI Coach

Provides suggested questions and a message area for asking about habit consistency, procrastination, missed days, motivation, morning routines, tracking progress, and other habit-related topics. The screenshot includes example responses encouraging small, repeatable actions.

See Figure 9 in Section 8.

6.8 Profile and settings

Allows the user to edit their name and timezone, select a theme, enable or disable daily reminders, save profile settings, and access a change-password form.

See Figure 10 in Section 8.

6.9 Motivational popup

Displays a short encouragement message in a modal dialog with close and Keep Going controls. The implementation notes describe keyboard dismissal and focus restoration support.

See Figure 11 in Section 8.

7. Data Management and Persistence

The backend uses Mongoose models and server routes to manage application data. Habit records are associated with users, and the activity model stores events for the authenticated user. Recorded activity types include habit creation, completion and undo, editing, deletion, pause and resume, daily-goal changes, and streak milestones. Activity entries can include snapshots of the habit name, category, and emoji so that the history remains understandable even if the habit is later changed.

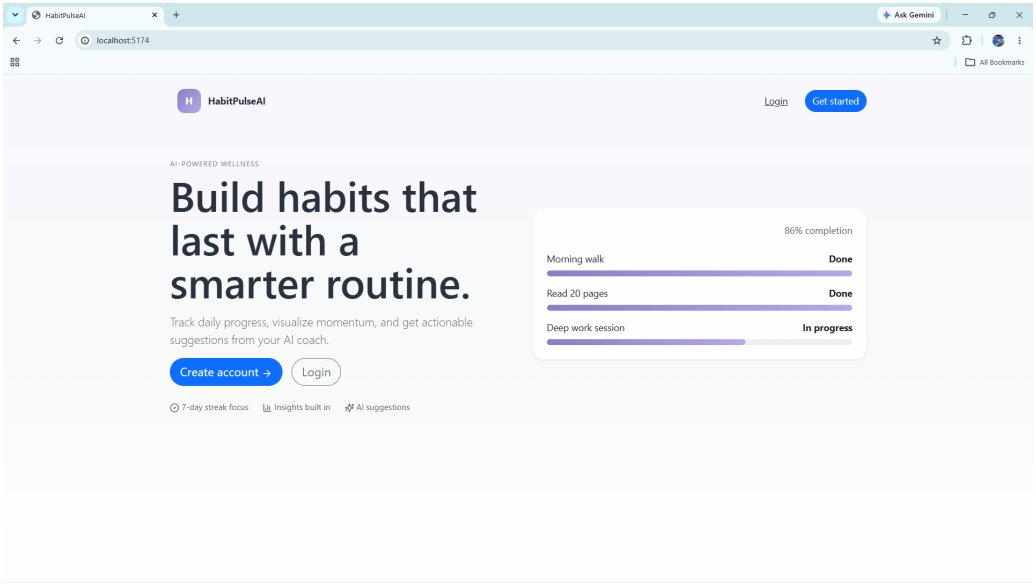
The implementation notes indicate that an activity collection and indexes are created through Mongoose; no manual database migration is required. Activity history begins when the feature is introduced and is not backfilled from older actions. The optional idempotency handling is intended to reduce duplicate activity records when requests are retried.

Persistence requirement: Persistent activity history requires MongoDB. If the server uses an in-memory fallback, those records are lost when the server stops.

8. Application Screenshots

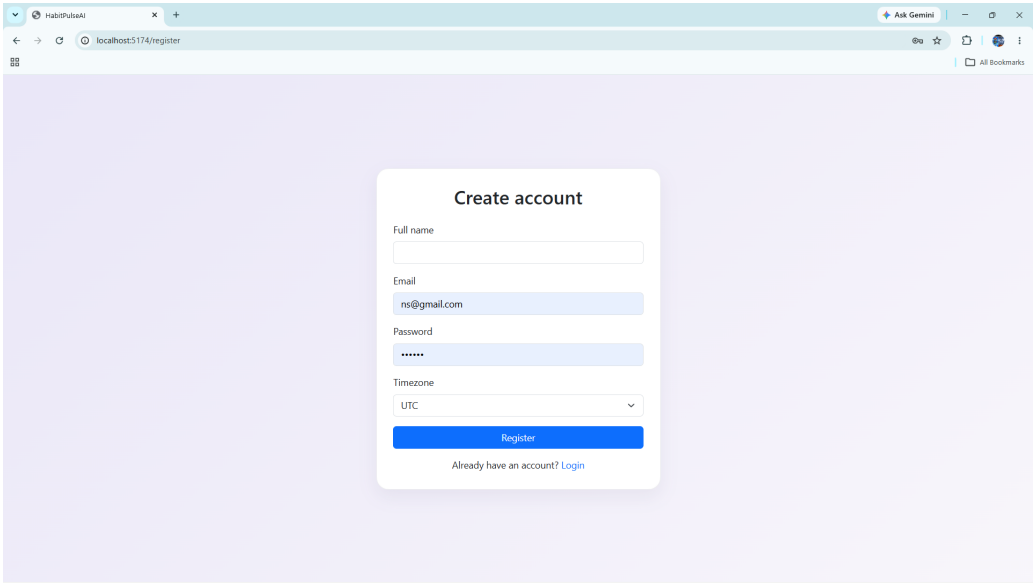
The following figures document the visible application pages supplied for this report. Screenshots retain their original colour to show the implemented interface and visual hierarchy.

Figure 1. HabitPulseAI landing page



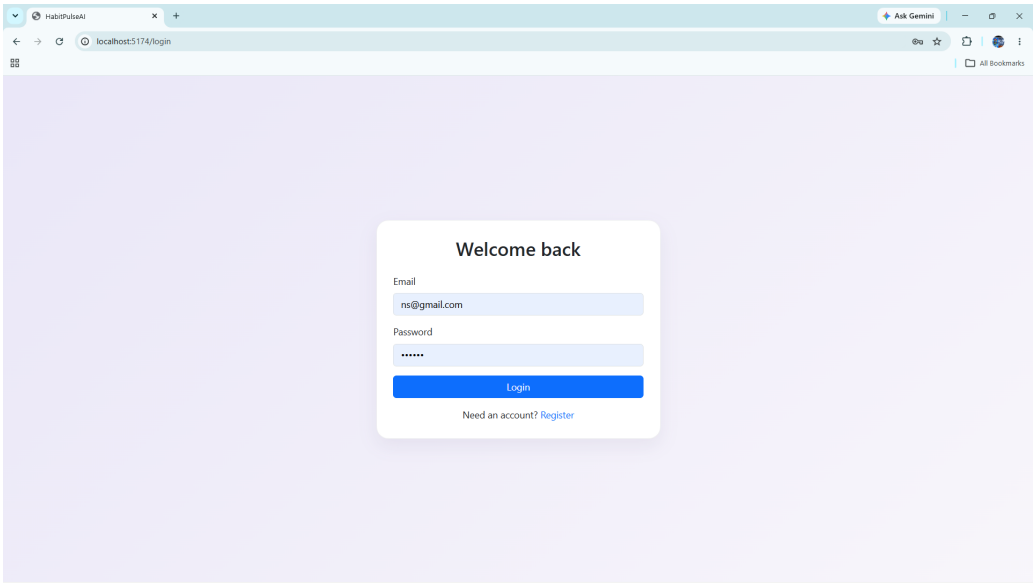
The landing page introduces the application and presents account access actions alongside a sample progress card.

Figure 2. Account registration



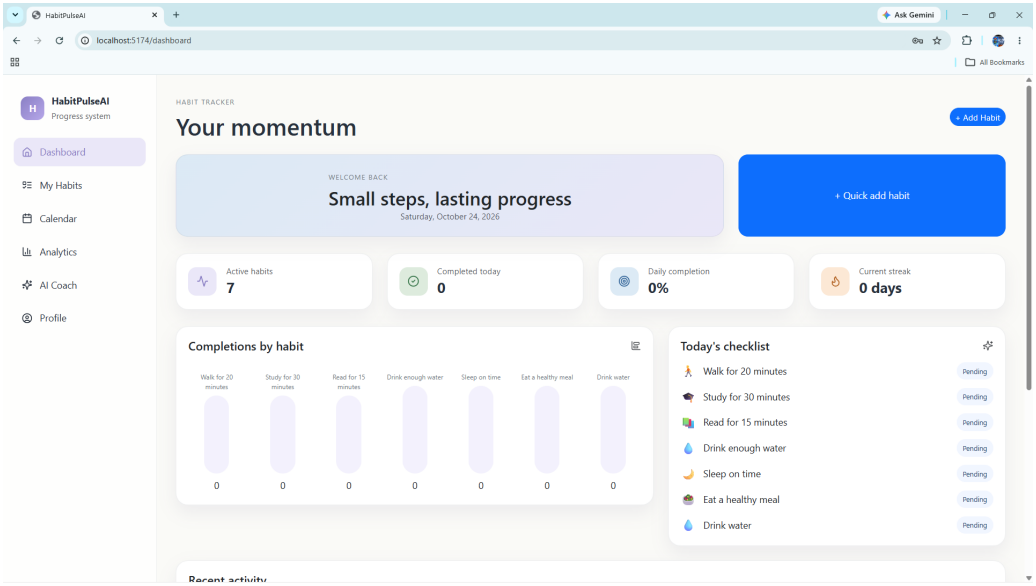
The registration form includes full name, email, password, and timezone fields.

Figure 3. User login



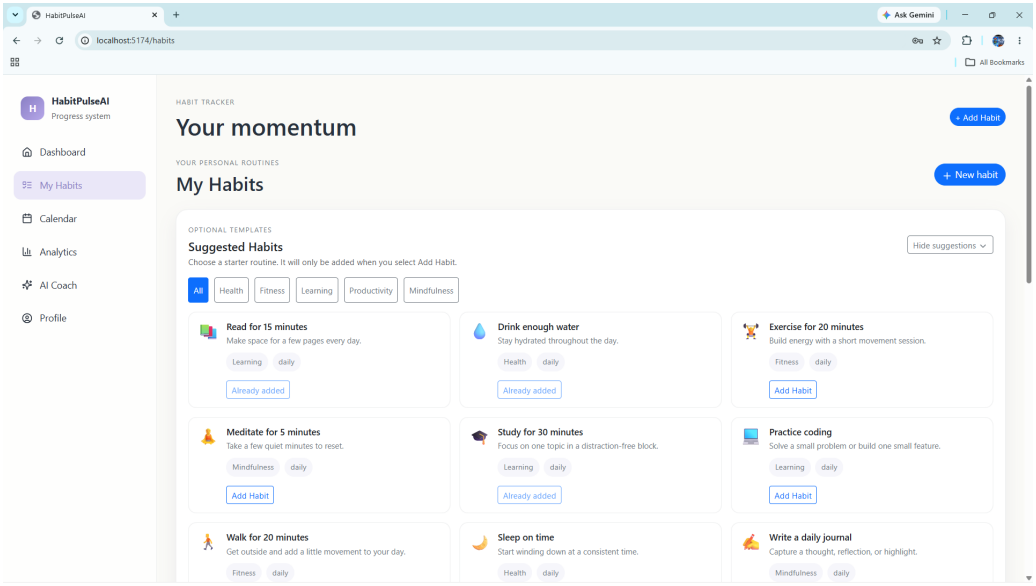
The login page provides email and password fields and a route to registration.

Figure 4. Dashboard overview



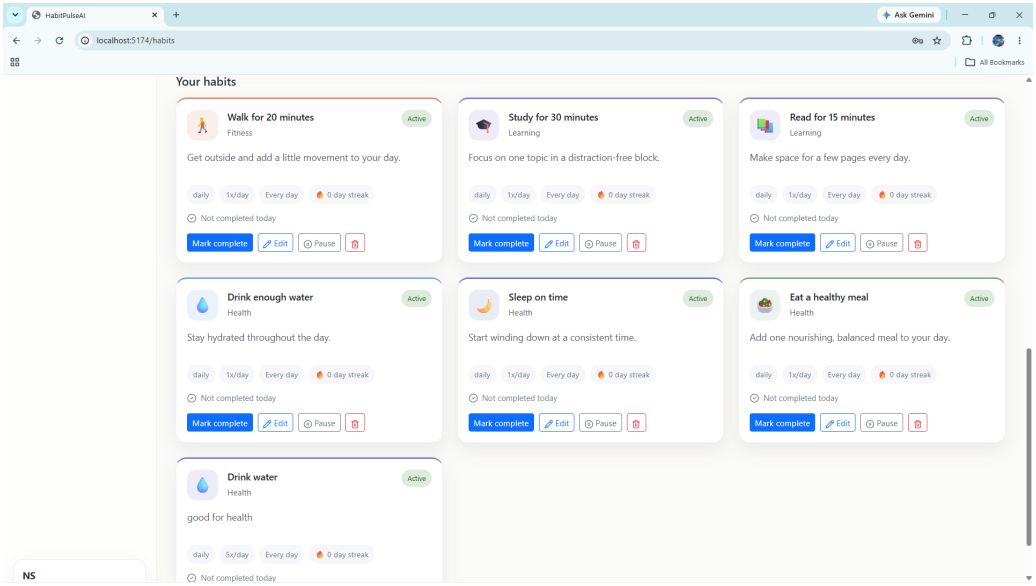
The dashboard summarises habits, daily completion metrics, a completion visualisation, and today's checklist.

Figure 5. Suggested habits



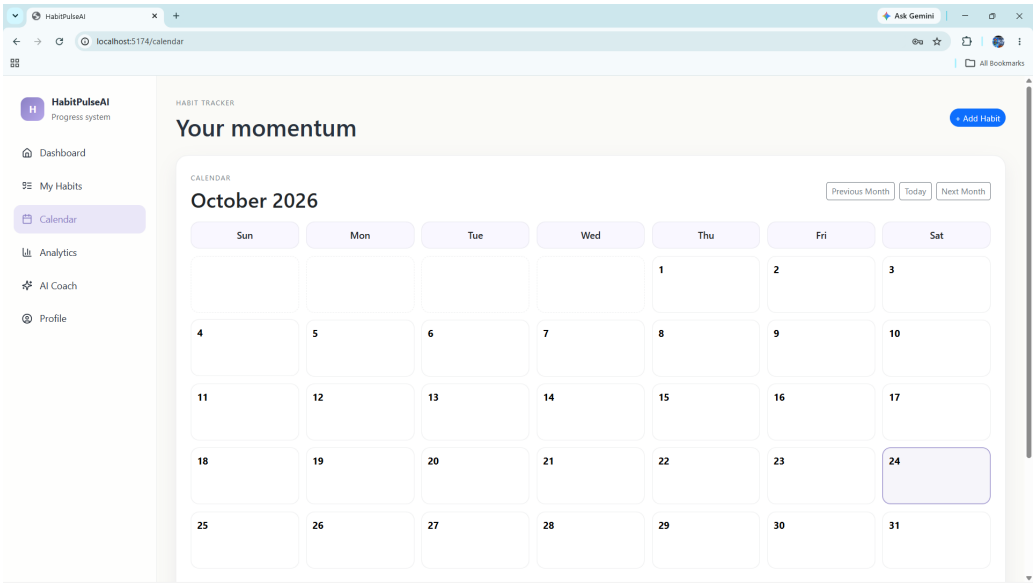
The My Habits page offers optional starter templates with category filters and Add Habit actions.

Figure 6. Habit cards and controls



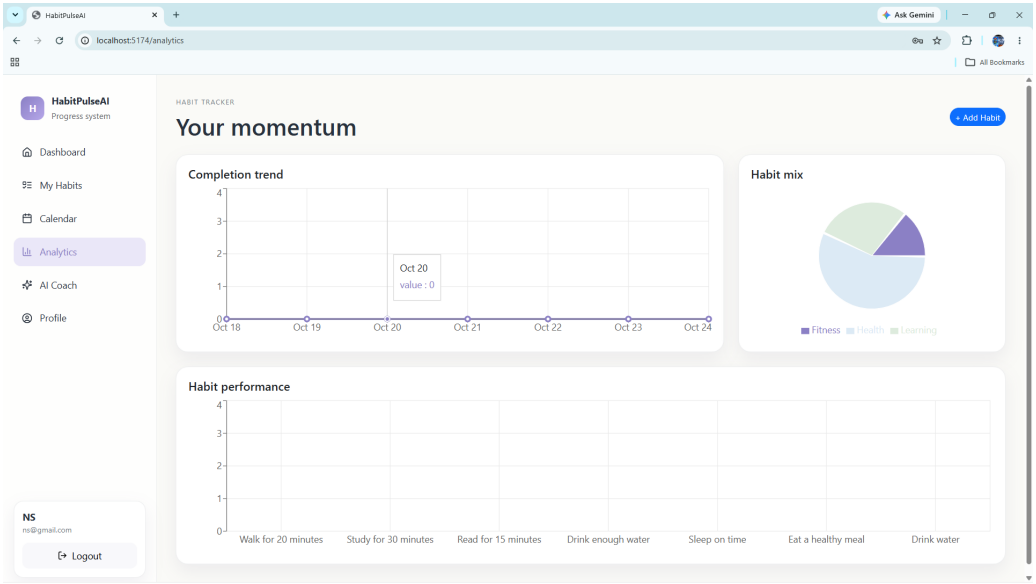
Habit cards display routine details and controls for completion, editing, pausing, and deletion.

Figure 7. Calendar view



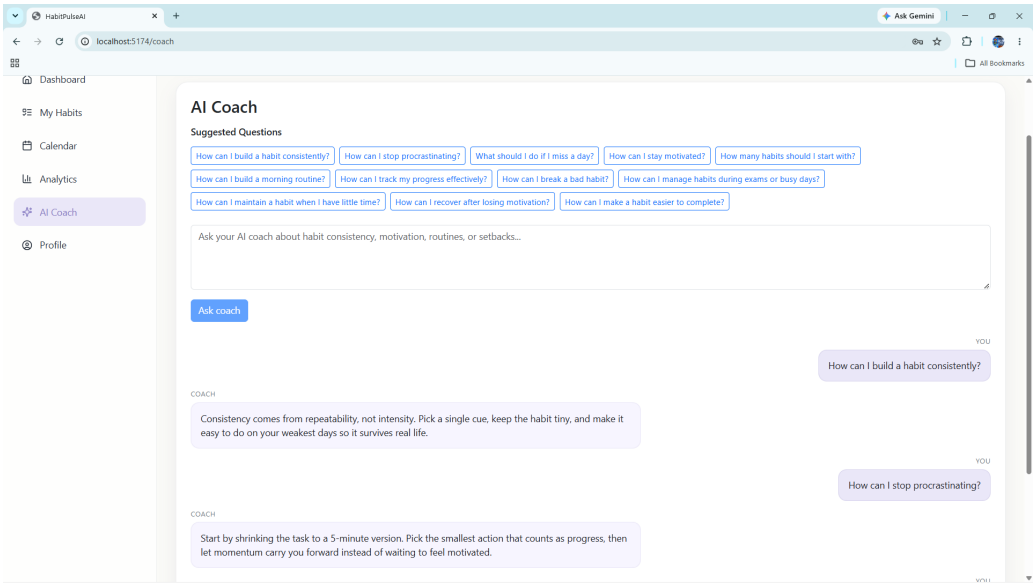
The calendar page includes month navigation and a date grid.

Figure 8. Analytics view



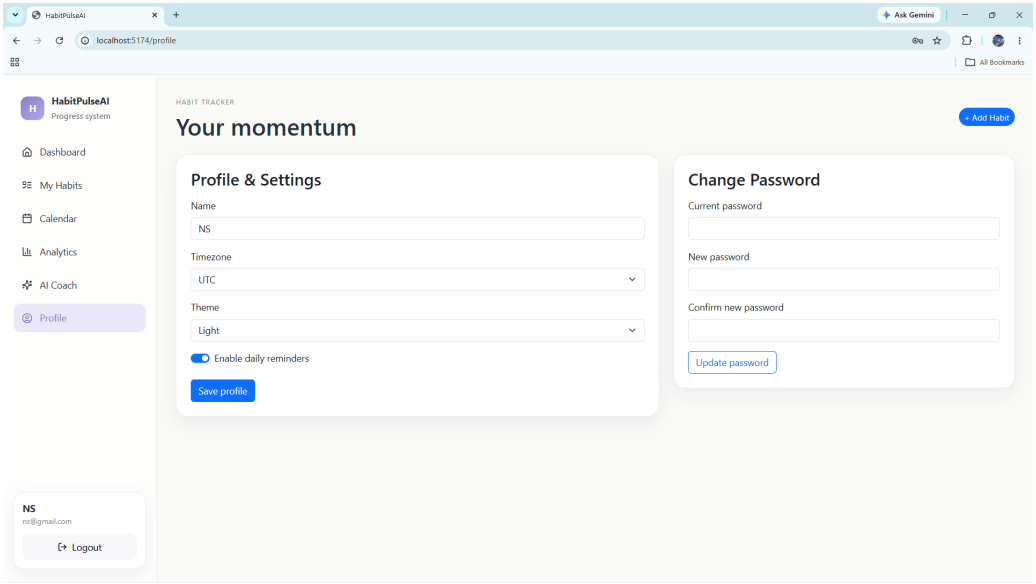
The analytics page displays completion trend, habit mix, and habit performance sections.

Figure 9. AI Coach



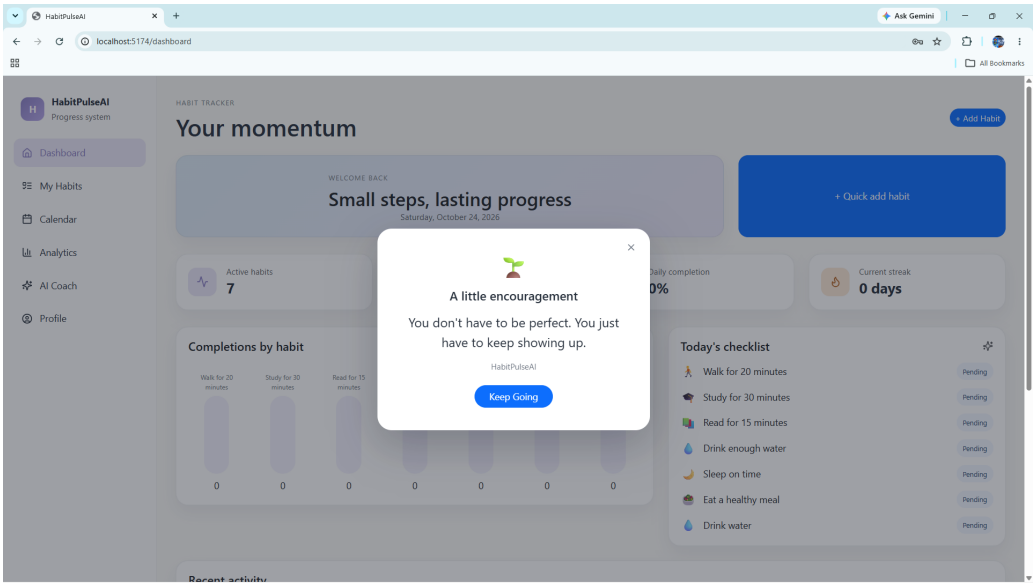
The AI Coach interface presents suggested questions and example coach responses.

Figure 10. Profile and settings



The profile page includes name, timezone, theme, reminders, and password update sections.

Figure 11. Motivational popup



A motivational dialog overlays the dashboard and offers a Keep Going action.

9. Testing and Verification

The project implementation notes report the following verification commands and results:

Area	Command	Reported result
Server tests	`npm --workspace server exec vitest run`	2 test files passed; 7 tests reported as passing.
Client tests	`npm --workspace client exec vitest run`	4 test files passed; 4 tests reported as passing.
Client build	`npm --workspace client run build`	Production build completed successfully.

The same notes mention non-fatal warnings: the production JavaScript bundle exceeds Vite's 500 kB advisory threshold, and React Router emits future-flag warnings.

Verification status: Automated tests and the build passed, but a full manual browser/device-width review was not reported as completed. Responsive layouts should therefore still be checked on actual narrow and wide viewports before release.

10. Limitations

- Activity history is not reconstructed for events that occurred before the activity feature was added.
- Persistent activity records depend on MongoDB being configured and available; in-memory fallback records do not survive a server restart.
- The screenshots show a sample account state and should not be interpreted as performance results from a long-term user study.
- Manual cross-device and browser-console verification was not confirmed in the implementation notes.
- The frontend build reports a bundle-size advisory that may affect initial loading on slower connections.

11. Future Enhancements

- Perform responsive testing on mobile, tablet, and desktop viewports and review accessibility with keyboard-only navigation and assistive technology.
- Use code splitting and lazy loading to reduce the initial JavaScript bundle size.
- Add optional export of habit and activity history for personal review.
- Provide clearer analytics summaries when there is sufficient completion history, while explaining empty or incomplete datasets.

- Add user-controlled reminder scheduling and verify reminder behaviour across timezones.
- Expand test coverage for habit lifecycle events, calendar date selection, analytics calculations, authentication boundaries, and activity persistence.
- Document environment variables and deployment steps without publishing credentials or secret values.

12. Conclusion

HabitPulseAI brings everyday habit management, calendar review, analytics, profile settings, motivational encouragement, and AI-assisted guidance into a single web application. The screenshots demonstrate the main user-facing pages, while the implementation notes describe supporting enhancements to responsive navigation, habit templates, habit cards, and persistent activity history. The reported automated tests and production build provide an initial verification baseline.

Further manual responsive and accessibility checks, along with attention to the bundle-size advisory, would strengthen the application before deployment.